

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ-1

DURATION: 35 MINUTES

SUMMER SEMESTER, 2023-2024

FULL MARKS: 15

Math 4641: Numerical Methods

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

1. Suppose, R and k are arbitrary numbers such that $R > 0$ and $k \neq 0$. Prove that the Newton-Raphson formula for finding the inverse of the k th root of R , which is mathematically denoted as $\frac{1}{\sqrt[k]{R}}$, is as shown in Equation 1. 3
(CO2)
(PO1)

$$x_{i+1} = \frac{1}{k} \left[(k-1)x_i + \frac{1}{Rx_i^{k-1}} \right] \quad (1)$$

Solution:

We know, the Newton-Raphson method formula for solving $f(x) = 0$ is,

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Suppose,

$$\begin{aligned} x &= \frac{1}{\sqrt[k]{R}} \\ \Rightarrow x^k &= \frac{1}{R} \\ \Rightarrow x^k - \frac{1}{R} &= 0 \end{aligned}$$

We take $f(x) = x^k - \frac{1}{R}$ and so $f'(x) = kx^{k-1}$.

Thus,

$$\begin{aligned} x_{i+1} &= x_i - \frac{x_i^k - \frac{1}{R}}{kx_i^{k-1}} \\ &= x_i - \frac{Rx_i^k - 1}{Rkx_i^{k-1}} \\ &= x_i - \left(\frac{Rx_i^k}{Rkx_i^{k-1}} - \frac{1}{Rkx_i^{k-1}} \right) \\ &= x_i - \left(\frac{x_i}{k} - \frac{1}{Rkx_i^{k-1}} \right) \\ &= \frac{kx_i - x_i}{k} + \frac{1}{Rkx_i^{k-1}} \\ &= \frac{(k-1)x_i}{k} + \frac{1}{Rkx_i^{k-1}} \\ &= \frac{1}{k} \left[\frac{(k-1)x_i}{k} + \frac{1}{Rx_i^{k-1}} \right] \quad \square \text{ Q.E.D.} \end{aligned}$$

Rubric:

- 1 point for using the Newton-Raphson approximation formula.
- 2 points for correct derivation.

2. a) What are significant digits? What is the relationship between the number of significant digits that are at least correct (m) in an approximation and the relative approximate error (ϵ_a)?

1
(CO2)
(PO1)

Solution:

Significant figures (also known as the significant digits, precision, or resolution) of a number in positional notation are digits in the number that are reliable and necessary to indicate the quantity of something. These are the digits that are known reliably in a reported number. The relationship between m and ϵ_a is,

$$|\epsilon_a| \leq 0.5 \times 10^{2-m}\%$$

Rubric:

- 0.5 point for a correct definition. (Doesn't need to be exactly as written above.)
- 0.5 point for the correct relationship.

- b) Determine the *minimum* number of significant digits in each of these numerical values —

3
(CO4)
(PO1)

(a) 69.0

(b) 69,420

(c) 6.94200×10^{911}

(d) 69.00420

(e) 069420

(f) 0.069420

Solution:

We need to determine the *minimum* number of significant digits.

- (i) 69.0 has 3 significant digits, since 0's to the right of the decimal point are significant.
- (ii) 69,420 has 4 significant digits, since trailing 0's in an integer may or may not be significant.
- (iii) 6.94200×10^{911} has 6 significant digits, since all digits of the significand/coefficient of a number's representation using Scientific Notation are significant.
- (iv) 69.00420 has 7 significant digits, since 0's between two significant non-zero digits are significant and trailing 0's in a number with the decimal point are significant.
- (v) 069420 has 4 significant digits, since leading 0's are not significant and trailing 0's in an integer may or may not be significant.
- (vi) 0.069420 has 5 significant digits, since leading 0's are not significant and trailing 0's in a number with the decimal point are significant.

Rubric:

- 0.5 point for each correct answer. (The reasons are not required.)

3. Draw a rough sketch of the graph of two functions $f_1(x)$ and $f_2(x)$ given that —

- For $f_1(x)$, False Position method performs better than the Bisection method.
- For $f_2(x)$, Bisection method performs better than the False Position method.

3
(CO3)
(PO2)

Here, “*performs better*” means that it takes a lesser number of iterations to converge to the root. You have the liberty to choose the initial guesses $\langle x_l, x_u \rangle$ as well.

Solution:

$f_1(x)$ may look something like,

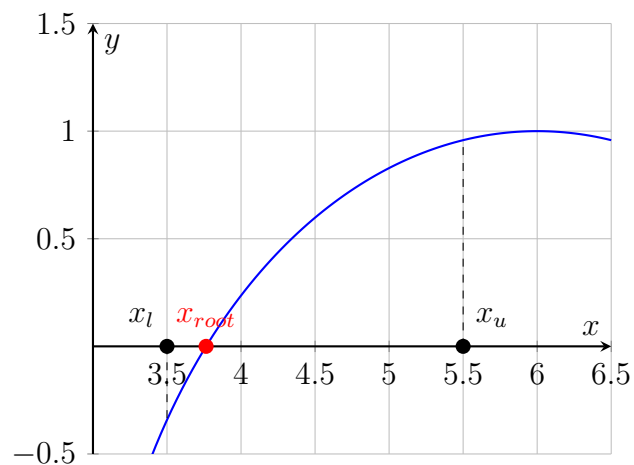


Figure 1: Graph of $f_1(x)$.

$f_2(x)$ may look something like,

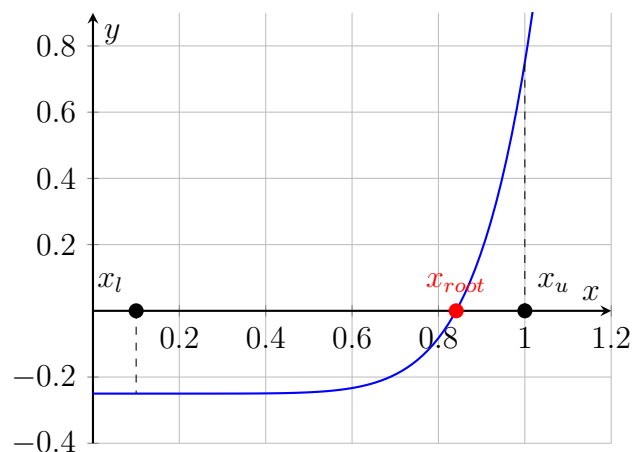


Figure 2: Graph of $f_2(x)$.

Rubric:

- 1.5 points for a correct $f_1(x)$ graph sketch.
- 1.5 points for a correct $f_2(x)$ graph sketch.

4. As depicted in Figure 3, the velocity of water, v (m/s), discharged from a cylindrical tank through a long pipe can be computed using Equation 2.

$$v = \sqrt{2gH} \tanh\left(\frac{\sqrt{2gH}}{2L}t\right) \quad (2)$$

where $g = 9.81 \text{ m/s}^2$ is the acceleration due to gravity, h is the initial height (m), L is the pipe length (m), and t is the elapsed time (s).

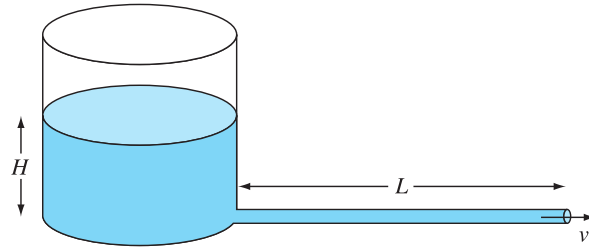


Figure 3: A cylindrical water tank for Question 4.

- a) Formulate a function $f(H)$ such that its root will yield the value of the height H for given values of v , L , and t .

1
(CO1)
(PO2)

Solution:

The given equation is,

$$v = \sqrt{2gH} \tanh\left(\frac{\sqrt{2gH}}{2L}t\right)$$

For planet Earth, the value of g is 9.81 m/s^2 , so the equation becomes,

$$v = \sqrt{19.62H} \tanh\left(\frac{\sqrt{19.62H}}{2L}t\right)$$

$$\Rightarrow v - \sqrt{19.62H} \tanh\left(\frac{\sqrt{19.62H}}{2L}t\right) = 0$$

So, the function $f(H)$ such that its root will yield the value of the height H for given values of v , L , and t is as follows:

$$f(H) = v - \sqrt{19.62H} \tanh\left(\frac{\sqrt{19.62H}}{2L}t\right)$$

- b) Apply the Secant method to estimate the height H that is needed to achieve $v = 5 \text{ m/s}$ in 2.5 s for a pipe of length 4 m . The initial guesses are $H_{-1} = 0$ and $H_0 = 2$. Demonstrating the step-by-step mathematical procedure, conduct 4 iterations, and find the relative approximate error ($|\epsilon_a|\%$) and the number of significant digits that are at least correct (m) at the end of each iteration. Draw Table 1 on your answer script and fill it out after performing the necessary calculations.

4
(CO2)
(PO1)

Solution:

After performing the series of calculations, we end up with the following table.

Iteration	H_{i-1}	H_i	H_{i+1}	$ \epsilon_a \%$	m	$f(H_{i+1})$
1	0	2	1.661	20.409	0	-0.396
2	2	1.661	1.446	14.88	0	0.0419
3	1.661	1.446	1.466	1.4	1	-0.0017
4	1.446	1.466	1.465	0.05	3	0

Table 1: The relevant values obtained in the answer of Question 4b

Iteration	H_{i-1}	H_i	H_{i+1}	$ \epsilon_a \%$	m	$f(H_{i+1})$
1	0	2				
2						
3						
4						